Вывод уравнения движения сплошной среды в координатах Эйлера

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За основу нашего вывода берём формулу (7) главы 2.9 теории упругости Новацкого, стр. 64.

$$\rho \cdot \frac{\mathrm{d}}{\mathrm{d}t} v_i = X_i + \sum_{j=1}^{3} \frac{\partial}{\partial \xi_j} \sigma_{ji}$$

В левой части материальная (местная) производная скорости материальной частицы по времени.

Для преобразования материальной производной используем уравнение (5) главы 2.8 той же книги, стр. 60

$$\frac{\mathrm{d}}{\mathrm{d}t} v_i(\xi, t) = \frac{\partial}{\partial t} v_i(\xi, t) + \sum_{j=1}^{3} v_j(\xi, t) \cdot \frac{\partial}{\partial \xi_j} v_i(\xi, t)$$

Здесь первый член правой части описывает изменение во времени функции $v_i(\xi, t)$ при постоянном ξ т.е. при неизменном положении в пространстве, а не для одной и той же материальной частицы. Второе слагаемое правой части называется конвективной производной. Здесь нужно иметь в виду, что формула (5) главы 2.8 Новацким получена "ограничиваясь величинами первого порядка малости по dt" разложения Тейлора от

$$\frac{v_i(\xi_j + v_j dt, t + dt) - v_i(\xi_j, t)}{dt}$$

То есть если ограничиться величинами хотя бы второго порядка малости по dt, то в уравнении движения сплошной среды появится ещё один источник нелинейности.

О скорости материальной частицы в описании Эйлера Новацкий пишет следующее: "В этом описании движение тела представляестя векторным полем $v_i(\xi_1, \xi_2, \xi_3, t)$, связанным с мгновенным положением материальной частицы." Однако, к сожалению, в работе Новацкого отсутсвует взаимосвязь скорости со смещением в описании Эйлера. Попробуем восполнить этот пробел. Для этого применим рассуждение Новацкого о вычислении материальной производной такой характеристики материальной среды как начальная координата частицы среды $x_i(\xi_1, \xi_2, \xi_3, t)$

$$v_i(\xi_1, \xi_2, \xi_3, t) = \frac{\mathrm{d}}{\mathrm{d}t} x_i(\xi_1, \xi_2, \xi_3, t) = \frac{\partial}{\partial t} x_i(\xi_1, \xi_2, \xi_3, t) + \sum_{j=1}^{3} v_j(\xi_1, \xi_2, \xi_3, t) \cdot \frac{\partial}{\partial \xi_j} x_i(\xi_1, \xi_2, \xi_3, t)$$

Первое слагаемое правой части представляет собой изменение во времени функции $x_i(\xi_1, \xi_2, \xi_3, t)$ (начальной координаты частиц среды) при неизменном положении в пространстве, а не для одной и той же материальной частицы. Но поскольку мы выводим уравнение движения сплошной среды, в основе которого лежит второй закон Ньютона, нам необходимо изменение во времени координат частиц среды именно для одной и той же материальной частицы. Именно в связи с этим возникает необходимость второго слагаемого: материальной (конвективной, субстанционной) производной.

Переносим конвективную производную в левую часть

$$v_i(\xi_1, \xi_2, \xi_3, t) - \sum_{j=1}^{3} v_j(\xi_1, \xi_2, \xi_3, t) \cdot \frac{\partial}{\partial \xi_j} x_i(\xi_1, \xi_2, \xi_3, t) = \frac{\partial}{\partial t} x_i(\xi_1, \xi_2, \xi_3, t)$$

Раскрываем знак суммы

$$v_i(\xi_1, \xi_2, \xi_3, t) - v_1(\xi_1, \xi_2, \xi_3, t) \cdot \frac{\partial}{\partial \xi_1} x_i(\xi_1, \xi_2, \xi_3, t) - v_2(\xi_1, \xi_2, \xi_3, t) \cdot \frac{\partial}{\partial \xi_2} x_i(\xi_1, \xi_2, \xi_3, t) - v_3(\xi_1, \xi_2, \xi_3, t) \cdot \frac{\partial}{\partial \xi_3} x_i(\xi_1, \xi_2, \xi_3, t) = \frac{\partial}{\partial t} x_i(\xi_1, \xi_2,$$

Раскрывая индекс i, переписываем в форме системы уравнений

$$v_1(\xi_1,\xi_2,\xi_3,t) - v_1(\xi_1,\xi_2,\xi_3,t)\cdot\frac{\partial}{\partial\xi_1}x_1(\xi_1,\xi_2,\xi_3,t) - v_2(\xi_1,\xi_2,\xi_3,t)\cdot\frac{\partial}{\partial\xi_2}x_1(\xi_1,\xi_2,\xi_3,t) - v_3(\xi_1,\xi_2,\xi_3,t)\cdot\frac{\partial}{\partial\xi_3}x_1(\xi_1,\xi_2,\xi_3,t) = \frac{\partial}{\partial t}x_1(\xi_1,\xi_2,$$

$$v_2(\xi_1,\xi_2,\xi_3,t) - v_1(\xi_1,\xi_2,\xi_3,t)\cdot\frac{\partial}{\partial\xi_1}x_2(\xi_1,\xi_2,\xi_3,t) - v_2(\xi_1,\xi_2,\xi_3,t)\cdot\frac{\partial}{\partial\xi_2}x_2(\xi_1,\xi_2,\xi_3,t) - v_3(\xi_1,\xi_2,\xi_3,t)\cdot\frac{\partial}{\partial\xi_3}x_2(\xi_1,\xi_2,\xi_3,t) = \frac{\partial}{\partial t}x_2(\xi_1,\xi_2,$$

$$v_3(\xi_1,\xi_2,\xi_3,t) - v_1(\xi_1,\xi_2,\xi_3,t)\cdot\frac{\partial}{\partial\xi_1}x_3(\xi_1,\xi_2,\xi_3,t) - v_2(\xi_1,\xi_2,\xi_3,t)\cdot\frac{\partial}{\partial\xi_2}x_3(\xi_1,\xi_2,\xi_3,t) - v_3(\xi_1,\xi_2,\xi_3,t)\cdot\frac{\partial}{\partial\xi_3}x_3(\xi_1,\xi_2,\xi_3,t) = \frac{\partial}{\partial t}x_3(\xi_1,\xi_2,$$

В матричной форме

$$\begin{bmatrix} 1-\frac{\partial}{\partial\xi_1}x_1(\xi_1,\xi_2,\xi_3,t) & -\frac{\partial}{\partial\xi_2}x_1(\xi_1,\xi_2,\xi_3,t) & -\frac{\partial}{\partial\xi_3}x_1(\xi_1,\xi_2,\xi_3,t) \\ -\frac{\partial}{\partial\xi_1}x_2(\xi_1,\xi_2,\xi_3,t) & 1-\frac{\partial}{\partial\xi_2}x_2(\xi_1,\xi_2,\xi_3,t) & -\frac{\partial}{\partial\xi_3}x_2(\xi_1,\xi_2,\xi_3,t) \\ -\frac{\partial}{\partial\xi_1}x_3(\xi_1,\xi_2,\xi_3,t) & -\frac{\partial}{\partial\xi_2}x_3(\xi_1,\xi_2,\xi_3,t) & 1-\frac{\partial}{\partial\xi_3}x_3(\xi_1,\xi_2,\xi_3,t) \end{bmatrix}\cdot\begin{bmatrix} v_1(\xi_1,\xi_2,\xi_3,t) \\ v_2(\xi_1,\xi_2,\xi_3,t) \\ v_3(\xi_1,\xi_2,\xi_3,t) \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial t}x_1(\xi_1,\xi_2,\xi_3, \\ \frac{\partial}{\partial t}x_2(\xi_1,\xi_2,\xi_3, \\ \frac{\partial}{\partial t}x_3(\xi_1,\xi_2,\xi_3, \end{bmatrix}$$

Выразим координату начальной точки среды через перемещения

Для описания Эйлера

$$x_i(\xi_1,\xi_2,\xi_3,t) = \xi_i - u_1(\xi_1,\xi_2,\xi_3,t)$$

Дифференцируем ее по координате Эйлера

$$\frac{\partial}{\partial\xi_j}x_i(\xi_1,\xi_2,\xi_3,t) = \delta_{ij} - \frac{\partial}{\partial\xi_j}u_i(\xi_1,\xi_2,\xi_3,t)$$

Дифференцируем по времени

$$\frac{\partial}{\partial t} x_i(\xi_1, \xi_2, \xi_3, t) = -\frac{\partial}{\partial t} u_i(\xi_1, \xi_2, \xi_3, t)$$

Подставляя в матричную систему уравнений, имеем

$$\begin{bmatrix} \frac{\partial}{\partial \xi_1} u_1(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_2} u_1(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_3} u_1(\xi_1, \xi_2, \xi_3, t) \\ \frac{\partial}{\partial \xi_1} u_2(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_2} u_2(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_3} u_2(\xi_1, \xi_2, \xi_3, t) \\ \frac{\partial}{\partial \xi_1} u_3(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_2} u_3(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_3} u_3(\xi_1, \xi_2, \xi_3, t) \end{bmatrix} \cdot \begin{bmatrix} v_1(\xi_1, \xi_2, \xi_3, t) \\ v_2(\xi_1, \xi_2, \xi_3, t) \\ v_3(\xi_1, \xi_2, \xi_3, t) \end{bmatrix} = \begin{bmatrix} -\frac{\partial}{\partial t} u_1(\xi_1, \xi_2, \xi_3, t) \\ -\frac{\partial}{\partial t} u_2(\xi_1, \xi_2, \xi_3, t) \\ -\frac{\partial}{\partial t} u_3(\xi_1, \xi_2, \xi_3, t) \end{bmatrix}$$

Решая систему, получаем выражение для скорости материальных частиц в зависимости от смещения в описании Эйлера

```
restart : with(linalg) :
```

$$vvv := linsolve\left(\begin{bmatrix} \frac{\partial}{\partial \xi_1} u_1(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_2} u_1(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_3} u_1(\xi_1, \xi_2, \xi_3, t) \\ \frac{\partial}{\partial \xi_1} u_2(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_2} u_2(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_3} u_2(\xi_1, \xi_2, \xi_3, t) \\ \frac{\partial}{\partial \xi_1} u_3(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_2} u_3(\xi_1, \xi_2, \xi_3, t) & \frac{\partial}{\partial \xi_3} u_3(\xi_1, \xi_2, \xi_3, t) \end{bmatrix}, \begin{bmatrix} -\frac{\partial}{\partial t} u_1(\xi_1, \xi_2, \xi_3, t) \\ -\frac{\partial}{\partial t} u_2(\xi_1, \xi_2, \xi_3, t) \\ -\frac{\partial}{\partial t} u_3(\xi_1, \xi_2, \xi_3, t) \end{bmatrix}\right) :$$

$$v_1(\xi_1, \xi_2, \xi_3, t) := vvv[1] :$$

$$v_2(\xi_1, \xi_2, \xi_3, t) := vvv[2] :$$

$$v_3(\xi_1, \xi_2, \xi_3, t) := vvv[3]:$$

Решение в общем виде, полученное Методом Крамера

$$v_1(\xi_1, \xi_2, \xi_3, t): \quad v_2(\xi_1, \xi_2, \xi_3, t): \quad v_3(\xi_1, \xi_2, \xi_3, t):$$

Проблема полученных формул, кроме их громоздкости, однако та, что если перемещения по одной или по двум координатам равны нулю, например в случае плоской волны (продольной или поперечной) то детерминант матрицы $\frac{\partial}{\partial \xi_j} u_i$ будет равен нулю, что приведёт к нулю в знаменателе. Поэтому в дальнной работе ниже представлена процедура вычисления скорости материальной частицы

Приступаем к непосредственному выводу уравнения движения

```
restart; with(Physics[Vectors]); Setup(mathematicalnotation = true) : with(LinearAlgebra) :
```

```
[&x, `+`, `.`, ChangeBasis, ChangeCoordinates, Component, Curl, DirectionalDiff, Divergence, Gradient, Identify, Laplacian, ∇, Norm, Setup,
```

координаты Эйлера

$$\xi := \xi_i \cdot _i + \xi_j \cdot _j + \xi_k \cdot _k : \xi$$

$$\xi_i \hat{i} + \xi_j \hat{j} + \xi_k \hat{k} \tag{2}$$

вектор перемещения

$$u := u_i(\xi_i, \xi_j, \xi_k, t) \cdot _i + u_j(\xi_i, \xi_j, \xi_k, t) \cdot _j + u_k(\xi_i, \xi_j, \xi_k, t) \cdot _k : u$$

$$u_i(\xi_i, \xi_j, \xi_k, t)\ \hat{i} + u_j(\xi_i, \xi_j, \xi_k, t)\ \hat{j} + u_k(\xi_i, \xi_j, \xi_k, t)\ \hat{k} \tag{3}$$

Заполняем матрицу производной смещения по координатам Эйлера

```
> du_dξ := Matrix(3):
  for i from 1 by 1 to 3
   do
     for j from 1 by 1 to 3
      do
        du_dξ[i,j] := diff(Component(u,i), Component(ξ,j))
      end do
   end do:
  du_dξ
```

$$\begin{bmatrix} \frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t) \\ \frac{\partial}{\partial \xi_i} u_j(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t) \\ \frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t) \end{bmatrix} \quad (4)$$

Вектор правых частей - производня смещения по времени

```
minus_du_dt := Vector(3): for i from 1 by 1 to 3 do minus_du_dt[i] := -diff(Component(u,i), t) end do: minus_du
```

$$\begin{bmatrix} -\left(\frac{\partial}{\partial t} u_i(\xi_i, \xi_j, \xi_k, t)\right) \\ -\left(\frac{\partial}{\partial t} u_j(\xi_i, \xi_j, \xi_k, t)\right) \\ -\left(\frac{\partial}{\partial t} u_k(\xi_i, \xi_j, \xi_k, t)\right) \end{bmatrix} \quad (5)$$

Определяем процедуру определения скорости материальных частиц в зависимости от смещения в описании Эйлера

```
calc_material_vel := proc(du_dξ, minus_du_dt) description "Calculate velocity of material particles in Eiler coordinates";
  vel := LinearSolve(du_dξ, minus_du_dt) :
  for i from 1 by 1 to 3 do
    if 0 = du_dξ[1, i] and 0 = du_dξ[2, i] and 0 = du_dξ[3, i] then
      vel := subs(vel[i] = 0, vel)
    fi
  end do:
  vel;

end proc:
```

$$v := calc_material_vel(du_d\xi, minus_du_dt) :$$

В общем случае

Для конкретного случая из общего выражения для скорости

$$LinearSolve\left(\begin{bmatrix} \frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t) \\ \frac{\partial}{\partial \xi_i} u_j(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t) \\ \frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t) & \frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t) \end{bmatrix}, \begin{bmatrix} -\left(\frac{\partial}{\partial t} u_i(\xi_i, \xi_j, \xi_k, t)\right) \\ -\left(\frac{\partial}{\partial t} u_j(\xi_i, \xi_j, \xi_k, t)\right) \\ -\left(\frac{\partial}{\partial t} u_k(\xi_i, \xi_j, \xi_k, t)\right) \end{bmatrix}\right)$$

Скорость среды в координатах Эйлера для плоской волны вычисляем более упрощённо

Например.

Для случая плоской продольной волны, распространяющейся вдоль оси X, получаем

$$LinearSolve\left(\begin{bmatrix} \frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} -\left(\frac{\partial}{\partial t} u_i(\xi_i, \xi_j, \xi_k, t)\right) \\ 0 \\ 0 \end{bmatrix}\right)$$

$du_d\xi_x_prodoln := Matrix(3) : du_d\xi_x_prodoln[1, 1] := diff\left(Component(u, 1), Component\left(\xi, 1\right)\right) : du_d\xi_x_proc$

$$\begin{bmatrix} \frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{6}$$

$minus_du_dt_x_prodoln := Vector(3) : minus_du_dt_x_prodoln[1] := -diff(Component(u, 1), t) : minus_du_dt_x_proc$

$$\begin{bmatrix} -\left(\frac{\partial}{\partial t} u_i(\xi_i, \xi_j, \xi_k, t)\right) \\ 0 \\ 0 \end{bmatrix} \tag{7}$$

$v_x_prodoln := calc_material_vel(du_d\xi_x_prodoln, minus_du_dt_x_prodoln)$

$$v_x_prodoln := \begin{bmatrix} -\dfrac{\dfrac{\partial}{\partial t} u_i(\xi_i, \xi_j, \xi_k, t)}{\dfrac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t)} \\ 0 \\ 0 \end{bmatrix} \tag{8}$$

Для случая плоской поперечной волны, распространяющейся вдоль X, колебания частиц среды в которой происходят вдоль оси Z

$$LinearSolve\left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \dfrac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -\left(\dfrac{\partial}{\partial t} u_k(\xi_i, \xi_j, \xi_k, t)\right) \end{bmatrix}\right)$$

$du_d\xi_x_poperech_z := Matrix(3) : du_d\xi_x_poperech_z[3, 1] := diff(Component(u, 3), Component(\xi, 1)) : du_d\xi_x_poperec$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \dfrac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) & 0 & 0 \end{bmatrix} \tag{9}$$

$minus_du_dt_x_poperech_z := Vector(3) : minus_du_dt_x_poperech_z[3] := -diff(Component(u, 3), t) : minus_du_dt_x_poperec$

$$\begin{bmatrix} 0 \\ 0 \\ -\left(\frac{\partial}{\partial t}\, u_k\left(\xi_i, \xi_j, \xi_k, t\right)\right) \end{bmatrix} \tag{10}$$

$v_x_poperech_z := calc_material_vel(du_d\xi_x_poperech_z, minus_du_dt_x_poperech_z)$

$$v_x_poperech_z := \begin{bmatrix} -\dfrac{\frac{\partial}{\partial t}\, u_k\left(\xi_i, \xi_j, \xi_k, t\right)}{\frac{\partial}{\partial \xi_i}\, u_k\left(\xi_i, \xi_j, \xi_k, t\right)} \\ 0 \\ 0 \end{bmatrix} \tag{11}$$

А это пример поперечной волны, распространяющейся вдоль оси Y, колебания частиц среды, в которой происходят вдоль оси X

$$calc_material_vel\left(\begin{bmatrix} 0 & -A\,k\sin\left(k\,\xi_j + \omega\,t + \delta\right) & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} A\,\omega\sin\left(k\,\xi_j + \omega\,t + \delta\right) \\ 0 \\ 0 \end{bmatrix}\right)$$

$$\begin{bmatrix} 0 \\ -\frac{\omega}{k} \\ 0 \end{bmatrix} \tag{12}$$

Объявляем матрицу для тензора Альманси

```
> η := Matrix(3, shape=symmetric) :
```

> $\eta[1,1] := \eta_{xx} : \eta[2,2] := \eta_{yy} : \eta[3,3] := \eta_{zz} :$

> $\eta[1,2] := \eta_{xy} : \eta[1,3] := \eta_{xz} : \eta[2,3] := \eta_{yz} :$

> η

$$\begin{bmatrix} \eta_{xx} & \eta_{xy} & \eta_{xz} \\ \eta_{xy} & \eta_{yy} & \eta_{yz} \\ \eta_{xz} & \eta_{yz} & \eta_{zz} \end{bmatrix} \tag{13}$$

Символ Кронекера

> *assume* $(m, integer)$: *assume*$(n, integer)$: delta $:= (m, n) \rightarrow$ **if** $m = n$ **then** 1 **else** 0 **fi**;

$$\delta := (m, n) \mapsto \textbf{if}\ m = n\ \textbf{then}\ 1\ \textbf{else}\ 0\ \textbf{end if} \tag{14}$$

Общее выражение для закона Гука - заполняем матрицу тензора напряжений

> $\sigma :=$ *Matrix*$(3,\ shape = symmetric)$: **for** i **from** 1 **by** 1 **to** 3 **do for** j **from** 1 **by** 1 **to** 3 **do** $\sigma[i,j] := \lambda \cdot \text{delta}(i,j) \cdot (\eta[1,1] + \eta[2,2] + \eta[3,3]) + 2 \cdot$
end do end do: σ

$$\begin{bmatrix} \lambda\left(\eta_{xx}+\eta_{yy}+\eta_{zz}\right)+2\,\mu\,\eta_{xx} & 2\,\mu\,\eta_{xy} & 2\,\mu\,\eta_{xz} \\ 2\,\mu\,\eta_{xy} & \lambda\left(\eta_{xx}+\eta_{yy}+\eta_{zz}\right)+2\,\mu\,\eta_{yy} & 2\,\mu\,\eta_{yz} \\ 2\,\mu\,\eta_{xz} & 2\,\mu\,\eta_{yz} & \lambda\left(\eta_{xx}+\eta_{yy}+\eta_{zz}\right)+2\,\mu\,\eta_{zz} \end{bmatrix} \quad (15)$$

Заполняем матрицу тензора Альманси

```
> for i from 1 by 1 to 3 do
  for j from 1 by 1 to 3 do

  η[i,j] := 1/2·(diff(Component(u,i),Component(ξ,j)) + diff(Component(u,j),Component(ξ,i)) − sum(diff(Component(u,k), Component
  · diff(Component(u,k),Component(ξ,j)),k=1..3))

  end do
  end do
```

$\eta[1,1]$

$$\frac{\partial}{\partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t) - \frac{\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i,\xi_j,\xi_k,t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_k(\xi_i,\xi_j,\xi_k,t)\right)^2}{2} \quad (16)$$

$\eta[1,2]$

$$\frac{\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i,\xi_j,\xi_k,t)\right)}{2} + \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i,\xi_j,\xi_k,t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i,\xi_j,\xi_k,t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial \xi_j} u_j(\xi_i,\xi_j,\xi_k,t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_k(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i,\xi_j,\xi_k,t)\right)}{2}$$

$\eta[1,3]$

$$\frac{\left(\frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} + \frac{\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2}$$

$\eta[2, 1]$

$$\frac{\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} + \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2}$$

$\eta[2, 2]$

$$\frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t) - \frac{\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} \quad (20)$$

$\eta[2, 3]$

$$\frac{\left(\frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2} + \frac{\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2}$$

$\eta[3, 1]$

$$\frac{\left(\frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} + \frac{\left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2}$$

$$-\frac{\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2}$$

$\eta[3, 2]$

$$\frac{\left(\frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2} + \frac{\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2}$$
$$- \frac{\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2}$$

$\eta[3, 3]$

$$\frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t) - \frac{\left(\frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} \quad (24)$$

Определяем процедуру заполнения тензора Альманси

```
fill_Almanci_tensor := proc(du_dξ) description "";
  η := Matrix(3, shape = symmetric) :
  for i from 1 by 1 to 3 do
    for j from 1 by 1 to 3 do
      # du_dξ[i,j] := diff(Component(u, i), Component(ξ, j))
      η[i,j] := 1/2 · (du_dξ[i,j] + du_dξ[j,i]) :
      for k from 1 by 1 to 3 do η[i,j] := η[i,j] + 1/2 · ( - du_dξ[k,i] · du_dξ[k,j] ) end do
    end do
  end do:
  η :
end proc:
```

```
> fill_Almanci_tensor(du_dξ)
```

$$\left[\left[\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)^2-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_j(\xi_i,\xi_j,\xi_k,t)\right)^2-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right)^2,\ \frac{1}{2}\frac{\partial}{\partial\xi_j}u_i(\xi_i,\xi_j,\xi_k,t)+\frac{1}{2}\frac{\partial}{\partial\xi_i}u_j(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_j}u_i(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_j(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_j}u_j(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_j}u_k(\xi_i,\xi_j,\xi_k,t)\right),\ \frac{1}{2}\frac{\partial}{\partial\xi_k}u_i(\xi_i,\xi_j,\xi_k,t)+\frac{1}{2}\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_i(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_j(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_j(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_k(\xi_i,\xi_j,\xi_k,t)\right)\right],$$

$$\left[\frac{1}{2}\frac{\partial}{\partial\xi_j}u_i(\xi_i,\xi_j,\xi_k,t)+\frac{1}{2}\frac{\partial}{\partial\xi_i}u_j(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_j}u_i(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_j(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_j}u_j(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_j}u_k(\xi_i,\xi_j,\xi_k,t)\right),\ \frac{\partial}{\partial\xi_j}u_j(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_j}u_i(\xi_i,\xi_j,\xi_k,t)\right)^2-\frac{1}{2}\left(\frac{\partial}{\partial\xi_j}u_j(\xi_i,\xi_j,\xi_k,t)\right)^2-\frac{1}{2}\left(\frac{\partial}{\partial\xi_j}u_k(\xi_i,\xi_j,\xi_k,t)\right)^2,\ \frac{1}{2}\frac{\partial}{\partial\xi_k}u_j(\xi_i,\xi_j,\xi_k,t)+\frac{1}{2}\frac{\partial}{\partial\xi_j}u_k(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_j}u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_i(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_j}u_j(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_j(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_j}u_k(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_k(\xi_i,\xi_j,\xi_k,t)\right)\right],$$

$$\left[\frac{1}{2}\frac{\partial}{\partial\xi_k}u_i(\xi_i,\xi_j,\xi_k,t)+\frac{1}{2}\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_i(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_j(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_j(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_k(\xi_i,\xi_j,\xi_k,t)\right),\ \frac{1}{2}\frac{\partial}{\partial\xi_k}u_j(\xi_i,\xi_j,\xi_k,t)+\frac{1}{2}\frac{\partial}{\partial\xi_j}u_k(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_j}u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_i(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_j}u_j(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_j(\xi_i,\xi_j,\xi_k,t)\right)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_j}u_k(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_k(\xi_i,\xi_j,\xi_k,t)\right),\ \frac{\partial}{\partial\xi_k}u_k(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_k}u_i(\xi_i,\xi_j,\xi_k,t)\right)^2-\frac{1}{2}\left(\frac{\partial}{\partial\xi_k}u_j(\xi_i,\xi_j,\xi_k,t)\right)^2-\frac{1}{2}\left(\frac{\partial}{\partial\xi_k}u_k(\xi_i,\xi_j,\xi_k,t)\right)^2\right]\right]$$

> *fill_Almanci_tensor(du_dξ_x_prodoln)*

$$\begin{bmatrix}\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)^2 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix} \tag{26}$$

> *fill_Almanci_tensor(du_dξ_x_poperech_z)*

$$\begin{bmatrix} -\frac{1}{2}\left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t)\right)^2 & 0 & \frac{1}{2}\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) \\ 0 & 0 & 0 \\ \frac{1}{2}\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) & 0 & 0 \end{bmatrix} \tag{27}$$

В матрицу тензора напряжений подставляем тензор Альманси

```
> σ := Matrix(3, shape = symmetric) :
  for i from 1 by 1 to 3 do
   for j from 1 by 1 to 3 do
  σ[i, j] := λ·delta(i, j)·(η[1, 1] + η[2, 2] + η[3, 3]) + 2·μ·η[i, j]
  end do
  end do
```

$\sigma[1, 1]$

$$\lambda\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t) - \frac{\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} + \frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t) - \frac{\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} + \frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t) - \frac{\left(\frac{\partial}{\partial \xi_k} u_i(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_k} u_j(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_k} u_k(\xi_i, \xi_j, \xi_k, t)\right)^2}{2}\right) + 2\mu\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t) - \frac{\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i, \xi_j, \xi_k, t)\right)^2}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t)\right)^2}{2}\right)$$

$\sigma[1, 2]$

$$2\mu\left(\frac{\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} + \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_i(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_j} u_i(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_j(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_j} u_j(\xi_i, \xi_j, \xi_k, t)\right)}{2} - \frac{\left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t)\right)\left(\frac{\partial}{\partial \xi_j} u_k(\xi_i, \xi_j, \xi_k, t)\right)}{2}\right)$$

$\sigma[1, 3]$

$$2\,\mu\left(\frac{\left(\frac{\partial}{\partial\xi_k}u_i(\xi_9,\xi_9,\xi_k,t)\right)}{2}+\frac{\left(\frac{\partial}{\partial\xi_9}u_k(\xi_9,\xi_9,\xi_k,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_i(\xi_9,\xi_9,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_i(\xi_9,\xi_9,\xi_k,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_j(\xi_9,\xi_9,\xi_k,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_k(\xi_9,\xi_9,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_k(\xi_9,\xi_9,\xi_k,t)\right)}{2}\right)$$

$\sigma[2, 1]$

$$2\,\mu\left(\frac{\left(\frac{\partial}{\partial\xi_9}u_i(\xi_9,\xi_9,\xi_k,t)\right)}{2}+\frac{\left(\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_i(\xi_9,\xi_9,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_9}u_i(\xi_9,\xi_9,\xi_k,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_k(\xi_9,\xi_9,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_9}u_k(\xi_9,\xi_9,\xi_k,t)\right)}{2}\right)$$

$\sigma[2, 2]$

$$\lambda\left(\frac{\partial}{\partial\xi_9}u_i(\xi_9,\xi_9,\xi_k,t)-\frac{\left(\frac{\partial}{\partial\xi_9}u_i(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_k(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}+\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)-\frac{\left(\frac{\partial}{\partial\xi_9}u_i(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_k(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}+\frac{\partial}{\partial\xi_k}u_k(\xi_9,\xi_9,\xi_k,t)-\frac{\left(\frac{\partial}{\partial\xi_k}u_i(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_k}u_j(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_k}u_k(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}\right)+2\,\mu\left(\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)-\frac{\left(\frac{\partial}{\partial\xi_9}u_i(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_k(\xi_9,\xi_9,\xi_k,t)\right)^2}{2}\right)$$

$\sigma[2, 3]$

$$2\,\mu\left(\frac{\left(\frac{\partial}{\partial\xi_k}u_j(\xi_9,\xi_9,\xi_k,t)\right)}{2}+\frac{\left(\frac{\partial}{\partial\xi_9}u_k(\xi_9,\xi_9,\xi_k,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_i(\xi_9,\xi_9,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_i(\xi_9,\xi_9,\xi_k,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_j(\xi_9,\xi_9,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_j(\xi_9,\xi_9,\xi_k,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_9}u_k(\xi_9,\xi_9,\xi_k,t)\right)\left(\frac{\partial}{\partial\xi_k}u_k(\xi_9,\xi_9,\xi_k,t)\right)}{2}\right)$$

$\sigma[3, 1]$

$$2\,\mu\left(\frac{\left(\frac{\partial}{\partial\xi_3}u_i(\xi_1,\xi_2,\xi_3,t)\right)}{2}+\frac{\left(\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_1}u_i(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial}{\partial\xi_3}u_i(\xi_1,\xi_2,\xi_3,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_1}u_j(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial}{\partial\xi_3}u_j(\xi_1,\xi_2,\xi_3,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial}{\partial\xi_3}u_k(\xi_1,\xi_2,\xi_3,t)\right)}{2}\right)$$

$\sigma[3, 2]$

$$2\,\mu\left(\frac{\left(\frac{\partial}{\partial\xi_3}u_j(\xi_1,\xi_2,\xi_3,t)\right)}{2}+\frac{\left(\frac{\partial}{\partial\xi_2}u_k(\xi_1,\xi_2,\xi_3,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_2}u_i(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial}{\partial\xi_3}u_i(\xi_1,\xi_2,\xi_3,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_2}u_j(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial}{\partial\xi_3}u_j(\xi_1,\xi_2,\xi_3,t)\right)}{2}-\frac{\left(\frac{\partial}{\partial\xi_2}u_k(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial}{\partial\xi_3}u_k(\xi_1,\xi_2,\xi_3,t)\right)}{2}\right)$$

$\sigma[3, 3]$

$$\lambda\left(\frac{\partial}{\partial\xi_2}u_i(\xi_1,\xi_2,\xi_3,t)-\frac{\left(\frac{\partial}{\partial\xi_2}u_i(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_2}u_j(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_2}u_k(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}+\frac{\partial}{\partial\xi_2}u_j(\xi_1,\xi_2,\xi_3,t)-\frac{\left(\frac{\partial}{\partial\xi_2}u_i(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_2}u_j(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_2}u_k(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}+\frac{\partial}{\partial\xi_3}u_k(\xi_1,\xi_2,\xi_3,t)-\frac{\left(\frac{\partial}{\partial\xi_3}u_i(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_3}u_j(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_3}u_k(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}\right)+2\,\mu\left(\frac{\partial}{\partial\xi_3}u_k(\xi_1,\xi_2,\xi_3,t)-\frac{\left(\frac{\partial}{\partial\xi_3}u_i(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_3}u_j(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}-\frac{\left(\frac{\partial}{\partial\xi_3}u_k(\xi_1,\xi_2,\xi_3,t)\right)^2}{2}\right)$$

Определяем процедуру вычисления тензора напряжений на основе тензора Альманси

```
fill_stress_tensor_with_Almanci_tensor := proc(η) description "";
  σ := Matrix(3, shape = symmetric) :
  for i from 1 by 1 to 3 do
    for j from 1 by 1 to 3 do
      σ[i, j] := λ·delta(i, j)·(η[1, 1] + η[2, 2] + η[3, 3]) + 2·μ·η[i, j]
    end do
  end do:
  σ :

end proc:
```

fill_stress_tensor_with_Almanci_tensor(*fill_Almanci_tensor*(*du_dξ*)) :

fill_stress_tensor_with_Almanci_tensor(*fill_Almanci_tensor*(*du_dξ_x_prodoln*))

$$\left[\left[\lambda\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)^2\right)+2\,\mu\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)^2\right),0,0\right],\right.$$
$$\left[0,\lambda\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)^2\right),0\right],$$
$$\left.\left[0,0,\lambda\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)-\frac{1}{2}\left(\frac{\partial}{\partial\xi_i}u_i(\xi_i,\xi_j,\xi_k,t)\right)^2\right)\right]\right]$$

fill_stress_tensor_with_Almanci_tensor(*fill_Almanci_tensor*(*du_dξ_x_poperech_z*))

$$\left[\begin{array}{ccc}
-\frac{1}{2}\lambda\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right)^2-\mu\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right)^2 & 0 & \mu\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right) \\
0 & -\frac{1}{2}\lambda\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right)^2 & 0 \\
\mu\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right) & 0 & -\frac{1}{2}\lambda\left(\frac{\partial}{\partial\xi_i}u_k(\xi_i,\xi_j,\xi_k,t)\right.
\end{array}\right.$$

Сила, действующая на элемент объёма

```
> F := Vector(3) :
  F[1] := diff(σ[1, 1], Component(ξ, 1)) + diff(σ[1, 2], Component(ξ, 2)) + diff(σ[1, 3], Component(ξ, 3)) :
  F[2] := diff(σ[2, 1], Component(ξ, 1)) + diff(σ[2, 2], Component(ξ, 2)) + diff(σ[2, 3], Component(ξ, 3)) :
  F[3] := diff(σ[3, 1], Component(ξ, 1)) + diff(σ[3, 2], Component(ξ, 2)) + diff(σ[3, 3], Component(ξ, 3)) :
```

$$F_ := simplify(F[1]\cdot _i + F[2]\cdot _j + F[3]\cdot _k) :$$

Определяем процедуру вычисления силы, действующей на элемент объёма, исходя из тензора напряжений

```
calc_force := proc(sigma) description "";
F := Vector(3) :
 F[1] := diff(σ[1, 1], Component(ξ, 1)) + diff(σ[1, 2], Component(ξ, 2)) + diff(σ[1, 3], Component(ξ, 3)) :
 F[2] := diff(σ[2, 1], Component(ξ, 1)) + diff(σ[2, 2], Component(ξ, 2)) + diff(σ[2, 3], Component(ξ, 3)) :
 F[3] := diff(σ[3, 1], Component(ξ, 1)) + diff(σ[3, 2], Component(ξ, 2)) + diff(σ[3, 3], Component(ξ, 3)) :
F;

end proc:
```

$$calc_force(fill_stress_tensor_with_Almanci_tensor(fill_Almanci_tensor(du_d\xi_x_prodoln)))$$

$$\left[\begin{array}{l} \lambda\left(\frac{\partial^2}{\partial \xi_i^2} u_i(\xi_i,\xi_j,\xi_k,t)\right) - \left(\frac{\partial}{\partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial^2}{\partial \xi_i^2} u_i(\xi_i,\xi_j,\xi_k,t)\right) + 2\mu\left(\frac{\partial^2}{\partial \xi_i^2} u_i(\xi_i,\xi_j,\xi_k,t) - \left(\frac{\partial}{\partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial^2}{\partial \xi_i^2} u_i(\xi_i,\xi_j,\xi_k,t)\right.\right. \\ \qquad \lambda\left(\frac{\partial^2}{\partial \xi_j \partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t) - \left(\frac{\partial}{\partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial^2}{\partial \xi_j \partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t)\right)\right) \\ \qquad \lambda\left(\frac{\partial^2}{\partial \xi_k \partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t) - \left(\frac{\partial}{\partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t)\right)\left(\frac{\partial^2}{\partial \xi_k \partial \xi_i} u_i(\xi_i,\xi_j,\xi_k,t)\right)\right) \end{array}\right.$$

$$calc_force(fill_stress_tensor_with_Almanci_tensor(fill_Almanci_tensor(du_d\xi_x_poperech_z)))$$

$$\left[-\left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) \right) \left(\frac{\partial^2}{\partial \xi_i^2} u_k(\xi_i, \xi_j, \xi_k, t) \right) \lambda - 2 \left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) \right) \left(\frac{\partial^2}{\partial \xi_i^2} u_k(\xi_i, \xi_j, \xi_k, t) \right) \mu + \mu \left(\frac{\partial^2}{\partial \xi_k \partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t \right. \right.$$

$$- \left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) \right) \left(\frac{\partial^2}{\partial \xi_j \partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) \right) \lambda$$

$$\left. \mu \left(\frac{\partial^2}{\partial \xi_i^2} u_k(\xi_i, \xi_j, \xi_k, t) \right) - \left(\frac{\partial}{\partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) \right) \left(\frac{\partial^2}{\partial \xi_k \partial \xi_i} u_k(\xi_i, \xi_j, \xi_k, t) \right) \lambda \right.$$

Уравнение движения с учётом конвективной производной покомпонентно

```
Eq := Vector(3) :
Eq[1] := ρ·diff(vel_x, t) + ρ·vel_x·diff(vel_x, ξ_i) + ρ·vel_y·diff(vel_x, ξ_j) + ρ·vel_z·diff(vel_x, ξ_k) = F[1] :
Eq[2] := ρ·diff(vel_y, t) + ρ·vel_x·diff(vel_y, ξ_i) + ρ·vel_y·diff(vel_y, ξ_j) + ρ·vel_z·diff(vel_y, ξ_k) = F[2] :
Eq[3] := ρ·diff(vel_z, t) + ρ·vel_x·diff(vel_z, ξ_i) + ρ·vel_y·diff(vel_z, ξ_j) + ρ·vel_z·diff(vel_z, ξ_k) = F[3] :
```

Введём функцию плотности

$$\rho_f := \rho_m(\xi_i, \xi_j, \xi_k, t)$$

$$\rho_f := \rho_m(\xi_i, \xi_j, \xi_k, t) \tag{41}$$

Определим процедуру формирования системы дифференциальных уравнений движения с учетом конвективной производной скорости материальных частиц а также включая уравнение непрерывности

```
motion_equation := proc (ρ, v_material, F) description "";
Eq := Vector(4) :
Eq[1] := ρ·diff(v_material[1], t) + ρ·v_material[1]·diff(v_material[1], ξ_i) + ρ·v_material[2]·diff(v_material[1], ξ_j) + ρ·v_material[3]
·diff(v_material[1], ξ_k) = F[1] :
Eq[2] := ρ·diff(v_material[2], t) + ρ·v_material[1]·diff(v_material[2], ξ_i) + ρ·v_material[2]·diff(v_material[2], ξ_j) + ρ·v_material[3]
·diff(v_material[2], ξ_k) = F[2] :
Eq[3] := ρ·diff(v_material[3], t) + ρ·v_material[1]·diff(v_material[3], ξ_i) + ρ·v_material[2]·diff(v_material[3], ξ_j) + ρ·v_material[3]
·diff(v_material[3], ξ_k) = F[3] :
Eq[4] := diff(ρ, t) + diff(ρ·v_material[1], ξ_i) + diff(ρ·v_material[2], ξ_j) + diff(ρ·v_material[3], ξ_k) = 0;
```

Eq

end proc:

Теперь сформируем систему дифференциальных уравнений для продольной волны

$$Eq_x_prodoln := motion_equation\left(\rho_f, calc_material_vel(du_d\xi_x_prodoln, minus_du_dt_x_prodoln), calc_force(fill_stress_tensor_with_Almanci_tensor(fill_Almanci_tensor(du_d\xi_x_prodoln)))\right)$$

$$Eq_x_prodoln := \left[\left[\rho_m(\xi_x,\xi_y,\xi_z,t)\left(\frac{\left(\frac{\partial}{\partial t}u_i(\xi_x,\xi_y,\xi_z,t)\right)\left(\frac{\partial^2}{\partial\xi_x\partial t}u_i(\xi_x,\xi_y,\xi_z,t)\right)}{\left(\frac{\partial}{\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)\right)^2}-\frac{\frac{\partial^2}{\partial t^2}u_i(\xi_x,\xi_y,\xi_z,t)}{\frac{\partial}{\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)}\right)\right.\right.$$
$$\rho_m(\xi_x,\xi_y,\xi_z,t)\left(\frac{\partial}{\partial t}u_i(\xi_x,\xi_y,\xi_z,t)\right)\left(\frac{\left(\frac{\partial}{\partial t}u_i(\xi_x,\xi_y,\xi_z,t)\right)\left(\frac{\partial^2}{\partial\xi_x^2}u_i(\xi_x,\xi_y,\xi_z,t)\right)}{\left(\frac{\partial}{\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)\right)^2}-\frac{\frac{\partial^2}{\partial\xi_x\partial t}u_i(\xi_x,\xi_y,\xi_z,t)}{\frac{\partial}{\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)}\right)$$
$$\left.\left.-\frac{}{\frac{\partial}{\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)}=\lambda\left(\frac{\partial^2}{\partial\xi_x^2}u_i(\xi_x,\xi_y,\xi_z,t)-\left(\frac{\partial}{\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)\right)\left(\frac{\partial^2}{\partial\xi_x^2}u_i(\xi_x,\xi_y,\xi_z,t)\right)\right)+2\mu\left(\frac{\partial^2}{\partial\xi_x^2}u_i(\xi_x,\xi_y,\xi_z,t)-\left(\frac{\partial}{\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)\right)\left(\frac{\partial^2}{\partial\xi_x^2}u_i(\xi_x,\xi_y,\xi_z,t)\right)\right)\right],$$

$$\left[0=\lambda\left(\frac{\partial^2}{\partial\xi_y\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)-\left(\frac{\partial}{\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)\right)\left(\frac{\partial^2}{\partial\xi_y\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)\right)\right)\right],$$

$$\left[0=\lambda\left(\frac{\partial^2}{\partial\xi_z\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)-\left(\frac{\partial}{\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)\right)\left(\frac{\partial^2}{\partial\xi_z\partial\xi_x}u_i(\xi_x,\xi_y,\xi_z,t)\right)\right)\right],$$

$$\left[\frac{\partial}{\partial t}\rho_m(\xi_1,\xi_2,\xi_3,t)-\frac{\left(\frac{\partial}{\partial\xi_1}\rho_m(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial}{\partial t}u_i(\xi_1,\xi_2,\xi_3,t)\right)}{\frac{\partial}{\partial\xi_1}u_i(\xi_1,\xi_2,\xi_3,t)}+\frac{\rho_m(\xi_1,\xi_2,\xi_3,t)\left(\frac{\partial}{\partial t}u_i(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial^2}{\partial\xi_1^2}u_i(\xi_1,\xi_2,\xi_3,t)\right)}{\left(\frac{\partial}{\partial\xi_1}u_i(\xi_1,\xi_2,\xi_3,t)\right)^2}\right.$$
$$\left.-\frac{\rho_m(\xi_1,\xi_2,\xi_3,t)\left(\frac{\partial^2}{\partial\xi_1\partial t}u_i(\xi_1,\xi_2,\xi_3,t)\right)}{\frac{\partial}{\partial\xi_1}u_i(\xi_1,\xi_2,\xi_3,t)}=0\right]\Bigg]$$

сформируем систему дифференциальных уравнений для поперечной волны

$$Eq_x_poperech_z := motion_equation\left(\rho_f, calc_material_vel(du_d\xi_x_poperech_z, minus_du_dt_x_poperech_z),\right.$$
$$\left.calc_force(fill_stress_tensor_with_Almanci_tensor(fill_Almanci_tensor(du_d\xi_x_poperech_z)))\right)$$

$$Eq_x_poperech_z := \left[\left[\rho_m(\xi_1,\xi_2,\xi_3,t)\left(\frac{\left(\frac{\partial}{\partial t}u_k(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial^2}{\partial\xi_1\partial t}u_k(\xi_1,\xi_2,\xi_3,t)\right)}{\left(\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)\right)^2}-\frac{\frac{\partial^2}{\partial t^2}u_k(\xi_1,\xi_2,\xi_3,t)}{\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)}\right)\right.\right.$$
$$\left.-\frac{\rho_m(\xi_1,\xi_2,\xi_3,t)\left(\frac{\partial}{\partial t}u_k(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\left(\frac{\partial}{\partial t}u_k(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial^2}{\partial\xi_1^2}u_k(\xi_1,\xi_2,\xi_3,t)\right)}{\left(\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)\right)^2}-\frac{\frac{\partial^2}{\partial\xi_1\partial t}u_k(\xi_1,\xi_2,\xi_3,t)}{\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)}\right)}{\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)}=-\left(\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial^2}{\partial\xi_1^2}\right.\right.$$
$$\left.\left.u_k(\xi_1,\xi_2,\xi_3,t)\right)\lambda-2\left(\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial^2}{\partial\xi_1^2}u_k(\xi_1,\xi_2,\xi_3,t)\right)\mu+\mu\left(\frac{\partial^2}{\partial\xi_1\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)\right)\right],$$
$$\left[0=-\left(\frac{\partial}{\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)\right)\left(\frac{\partial^2}{\partial\xi_1\partial\xi_1}u_k(\xi_1,\xi_2,\xi_3,t)\right)\lambda\right],$$

$$\left[\begin{array}{l} 0=\mu\left(\frac{\partial^2}{\partial\xi_j^{\,2}}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)\right)-\left(\frac{\partial}{\partial\xi_j}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)\right)\left(\frac{\partial^2}{\partial\xi_k\partial\xi_j}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)\right)\lambda \\ \frac{\partial}{\partial t}\,\rho_m\left(\xi_i,\xi_j,\xi_k,t\right)-\frac{\left(\frac{\partial}{\partial\xi_j}\,\rho_m\left(\xi_i,\xi_j,\xi_k,t\right)\right)\left(\frac{\partial}{\partial t}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)\right)}{\frac{\partial}{\partial\xi_j}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)}+\frac{\rho_m\left(\xi_i,\xi_j,\xi_k,t\right)\left(\frac{\partial}{\partial t}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)\right)\left(\frac{\partial^2}{\partial\xi_j^{\,2}}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)\right)}{\left(\frac{\partial}{\partial\xi_j}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)\right)^2} \\ -\frac{\rho_m\left(\xi_i,\xi_j,\xi_k,t\right)\left(\frac{\partial^2}{\partial\xi_j\partial t}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)\right)}{\frac{\partial}{\partial\xi_j}\,u_k\left(\xi_i,\xi_j,\xi_k,t\right)}=0 \end{array}\right]$$

Полученные уравнения носят явно нелинейный характер

Чтобы в этом убедиться попробуем подставить синусоиду в систему уравнений для плоской продольной волны

$$simplify\left(subs\left(u_i\left(\xi_i,\xi_j,\xi_k,t\right)=A\cdot\cos\left(\omega\cdot t+k\cdot\xi_i+\delta\right),Eq_x_prodoln\right)\right)$$

$$\left[\begin{array}{c} 0=-2\cos\left(k\,\xi_i+\omega\,t+\delta\right)\left(\mu+\frac{\lambda}{2}\right)A\left(A\,k\sin\left(k\,\xi_i+\omega\,t+\delta\right)+1\right)k^2 \\ 0=0 \\ 0=0 \\ \frac{\partial}{\partial t}\,\rho_m\left(\xi_i,\xi_j,\xi_k,t\right)-\frac{\left(\frac{\partial}{\partial\xi_i}\,\rho_m\left(\xi_i,\xi_j,\xi_k,t\right)\right)\omega}{k}=0 \end{array}\right] \qquad (44)$$

попробуем подставить синусоиду в систему уравнений для плоской поперечной волны

$$simplify\left(subs\left(u_k\left(\xi_i, \xi_j, \xi_k, t\right) = A\cdot\cos\left(\omega\cdot t + k\cdot\xi_i + \delta\right), Eq_x_poperech_z\right)\right)$$

$$\begin{bmatrix} 0 = -A^2\sin\left(k\,\xi_i + \omega\,t + \delta\right)\cos\left(k\,\xi_i + \omega\,t + \delta\right)k^3\left(\lambda + 2\,\mu\right) \\ 0 = 0 \\ 0 = -\mu\,A\,k^2\cos\left(k\,\xi_i + \omega\,t + \delta\right) \\ \frac{\partial}{\partial t}\,\rho_m\left(\xi_i, \xi_j, \xi_k, t\right) - \frac{\left(\frac{\partial}{\partial \xi_i}\,\rho_m\left(\xi_i, \xi_j, \xi_k, t\right)\right)\omega}{k} = 0 \end{bmatrix} \quad (45)$$

И что же невооружённым глазом видно что синусоида не удовлетворяет уравнению плоской волны: ни продольной ни поперечной. Что говорит о нелинейности действительного уравнения движения сплошной среды, в отличие от уравнения Ламе, вывод которого даётся в курсах линейной теории упругости.